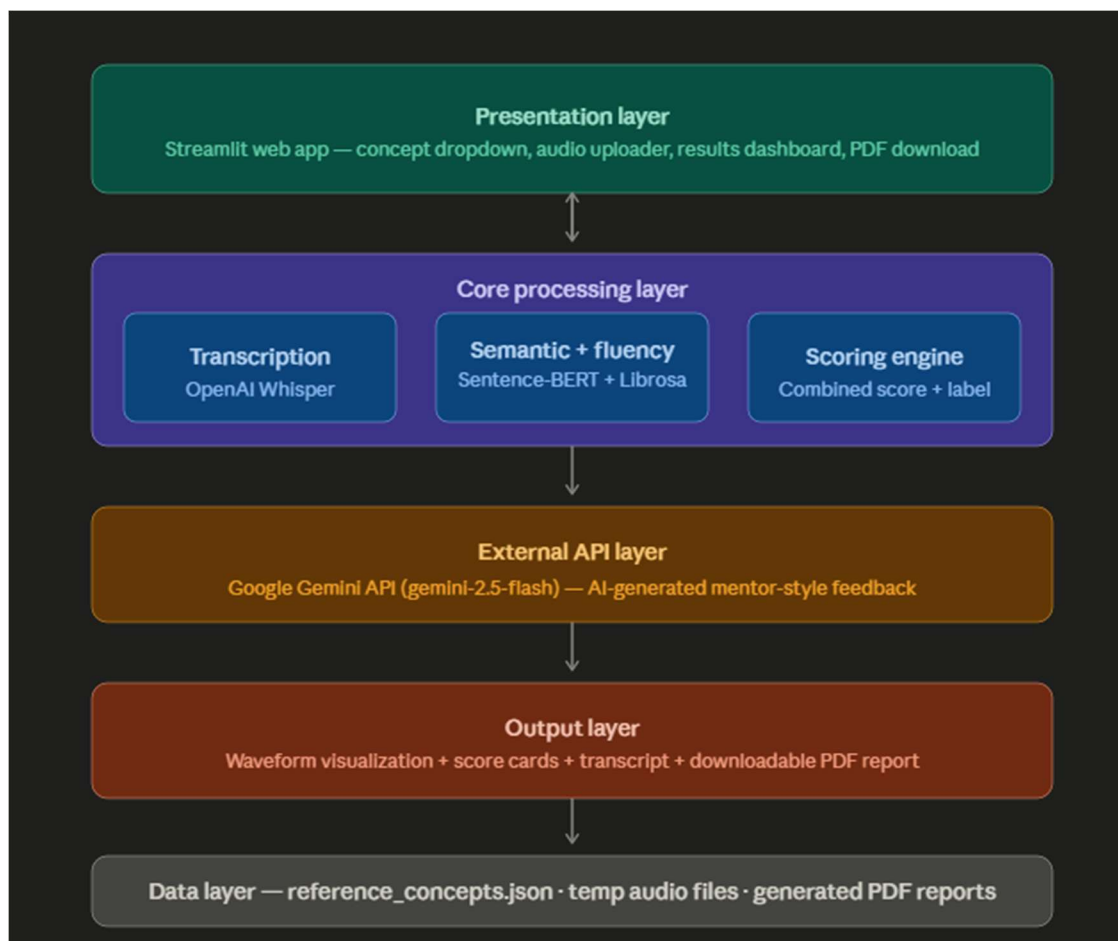


Solution Architecture

Date	28 June 2026
Team ID	
Project Name	Voice-Based Concept Understanding Analyser (VBCUA)
Maximum Marks	5 Marks

Solution Architecture Diagram:

Provide a high-level visual representation of your solution's architecture. Use the structural layout below as a template to map out your Presentation Layer, Application Layer, and Data Layer. Fill in the specific technologies and components for your project directly into the boxes to create your architectural diagram.



Component Description Table

Component Name	Description / Role in Architecture	Technologies Used
Presentation Layer	<i>Browser-based single-page UI where users upload audio, select a concept, trigger analysis, and view/download results</i>	Streamlit, HTML/CSS (custom dark theme)
Core Logic Service	Processes audio through the full AI pipeline — transcription, semantic scoring, fluency analysis, combined scoring, and AI feedback	<i>OpenAI Whisper, Sentence-BERT, Librosa, Python, Gemini API</i>
External API Integration	Generates natural-language mentor-style feedback from raw evaluation scores	<i>Google Gemini API (gemini-2.5-flash)</i>
<i>Data / Storage Layer</i>	Stores reference concept explanations (JSON), handles temporary audio uploads, and saves generated PDF reports	JSON file, Python tempfile, fpdf2